APEX RETAIL GROUP | PRACTICAL CLASS TASKS

DOCUMENT: CLASS TASKS (SESSION 3)

SCENARIO: REGIONAL STORE CLUSTER PERFORMANCE AUDIT

INSTRUCTIONS FOR LEARNER: Complete the following four tasks in your Jupyter Notebook during class. Follow the structured workflow: **Array Setup -> Vectorized Computation -> Dispersion Analysis -> Business Interpretation**. Do not peek at instructor solution keys until completed.

TASK 1: MATRIX CONSTRUCTS & SHAPE INSPECTION

Elena Rostova (Sales Director) provides revenue figures (in thousands of dollars) for 3 store clusters across 3 months (January, February, March):

- Cluster A: $120.0k, $135.0k, $128.0k
- Cluster B: $90.0k, $85.0k, $92.0k
- Cluster C: $210.0k, $225.0k, $218.0k

1. Import `numpy` as `np`.
2. Construct a 2D NumPy array named `cluster_revenue` representing this 3x3 matrix.
3. Print the array, its `.shape`, and its data type (`.dtype`).

TASK 2: VECTORIZED OPERATIONS & TARGET VARIANCES

Apex Retail set a monthly revenue target benchmark of $110.0k per cluster.

1. Create a variable `monthly_target = 110.0`.
2. Perform a vectorized subtraction to calculate `target_variance = cluster_revenue - monthly_target`.
3. Compute total monthly revenue across all clusters (sum down columns, `axis=0`).
4. Compute total 3-month revenue for each individual cluster (sum across rows, `axis=1`).
5. Print `target_variance`, total monthly revenues, and individual cluster totals.

TASK 3: MEASURING DISPERSION & BOOLEAN MASKING

In analysis, measuring the spread (dispersion) of revenue is as important as measuring the mean.

1. Calculate the overall average revenue across all clusters and months using `np.mean()`.
2. Calculate the standard deviation across all values using `np.std()`.
3. Use Boolean masking to identify which specific elements in `cluster_revenue` fell below the $110.0k target benchmark (`below_target_mask = cluster_revenue < monthly_target`).
4. Use the mask to extract and print the exact values that failed to reach the target.

TASK 4: EXECUTIVE INTERPRETATION & ADVISORY MEMO

Review your numerical outputs from Tasks 2 and 3:

1. Which specific cluster failed to hit the $110.0k target in every single month?
2. Compare Cluster B's 3-month total revenue against Cluster C's 3-month total revenue.
3. Draft a 2-3 sentence executive memo to Elena Rostova explaining how sales dispersion impacts regional targets.